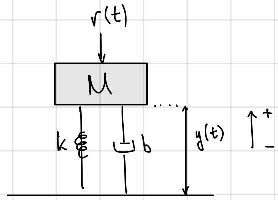
ELEC 3CL4 - Midterm Review Session w/ Dr. Davidson

* ensure to be clear about steps
  ↳ few words between each step

* if answer questionable, say its questionable
  ↳ if algebra / calc mistake then part marks '◡'

Q1) Mass spring damper question



• FBD :

• $\vec{F}_S = -k\underbrace{(y(t) - l_r)}$ = -ve when compressed = positive $\vec{F}$ going upwards

• $\vec{F}_D = -b\,y'(t)$ = damper squished = +ve $\vec{F}$

• $\vec{F}_g = Mg$

• $\sum F = m\vec{a}$
$= M\,y''(t)$

$F_S + F_D - F_G - r(t) = M\,y''(t)$

$M y''(t) - F_S - F_D = -F_G - r(t)$

$M y''(t) - (-k(y(t) - l_r)) - (-b y'(t)) = -Mg - r(t)$

$M y''(t) + k y(t) - k l_r + b y'(t) = -Mg - r(t)$

$\underbrace{M y''(t) + b y'(t) + k y(t)}_{\text{output}} = \underbrace{k l_r - Mg - r(t)}_{\text{input}}$ of system

$M s^2 Y(s) + b s Y(s) + k Y(s) = \cancel{\frac{k l_r}{s}} - \cancel{\frac{Mg}{s}} - R(s)$, $\frac{1}{s}$ = step function
, only care about relation between R & Y
↳ applied superposition

$Y(s)\left(Ms^2 + bs + k\right) = -R(s)$

$T(s) = \frac{Y(s)}{R(s)} = \frac{-1}{Ms^2 + bs + k}$

b) Steady State?

• Steady State = M not moving anymore → $\vec{a}$ & $\vec{v} = 0$

$M \cancelto{0}{y''(t)} + b \cancelto{0}{y'(t)} + k y(t) = k l_r - Mg - r(t)$

$k y(t) = k l_r - Mg - r(t)$

$y(t) = l_r - \frac{1}{k} Mg - \frac{1}{k} r(t)$

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* did 2025 midterm